



CSE342: Agile Software Development

Phase 1

Submitted to:

Dr. Mohamed Hassan Mahmoud El Gazzar

Submitted by:

Mohamed Abo Samra - 23P0016
Mai Hamed Hussein Abostate - 23P0261
Basmala Hany Mohamed - 23P0008
Mariam Shaker Zaki - 23P0312
Mariam Riyad Freeg - 23P0147
Maryam Hamdy Hassan - 23P0260

Date of submission:

17th April 2026

1. Agile Framework and Sprint Planning Approach

For Phase 1 of the project, the team followed the Scrum framework to plan and organize the development process for a single sprint without implementation.

1.1. Team Roles

The team members were assigned Scrum roles as follows:

- **Product Owner (PO) – Mai Hamed:** Responsible for defining and prioritizing the Product Backlog.
- **Scrum Master (SM) – Mohamed Abo Samra:** Responsible for facilitating Scrum events and ensuring adherence to Agile practices.
- **Development Team – Rest of the team:** Responsible for analyzing requirements, estimating effort, and preparing backlog items for implementation.

1.2. Scrum Events Planning

Although no implementation is required in Phase 1, all Scrum ceremonies were planned and reflected within Jira:

- **Sprint Planning:**
Conducted to define the Sprint 1 goal and select the Product Backlog Items (PBIs) included in the MVP.
- **Daily Scrum (Planned):**
Short daily meetings were planned to track progress, identify blockers, and ensure team alignment.
- **Sprint Review (Planned):**
Intended to demonstrate the completed MVP features and gather stakeholder feedback.
- **Sprint Retrospective (Planned):**
Planned to evaluate team performance and identify areas for improvement in future sprints.

1.3. Jira Usage and Agile Practices

All project activities were documented and managed using Jira in accordance with Agile best practices:

- Product Backlog Items were written using the standard user story format:
As a [role], I want [action], so that [benefit]
- Each backlog item includes:
 - Acceptance Criteria
 - Priority (MoSCoW)
 - Story Points estimation
 - Assigned team member

- Sprint 1 was created and configured on Jira, with all selected MVP items properly tracked and updated.

This approach ensures that the project follows a structured Agile methodology and is fully prepared for future implementation phases.

1.4. Sprint Goal

The goal of Sprint 1 is to deliver a usable and well-structured Minimum Viable Product (MVP) that provides essential functionality for university stakeholders while ensuring system stability and usability.

This sprint focuses on implementing core features that form the foundation of the system, including secure user authentication with role-based access control, basic user profile visibility, and access to key university information such as announcements and course catalog browsing. These features are carefully selected to ensure that users can interact with the system in a meaningful way from the very first release.

In addition, the sprint aims to establish a clear system structure and architecture that supports scalability and future development. By prioritizing simplicity and reliability, the team ensures that the delivered product is not only functional but also easy to test, evaluate, and extend in subsequent sprints.

By the end of Sprint 1, the system should demonstrate a complete user flow starting from login to basic navigation, providing immediate value to users. It will also serve as a strong baseline for integrating more advanced features such as messaging, enrollment management, and facilities handling in future iterations.

2. MVP Definition

The Minimum Viable Product (MVP) for Sprint 1 represents the smallest functional version of the university platform that delivers core value to users while remaining fully usable. The MVP focuses on three essential pillars:

- Secure access to the system through authentication and role-based control
- Basic user identity and profile visibility
- Communication of university-wide announcements and browsable content

The MVP deliberately excludes advanced features such as messaging systems and facility management, which are deferred to future sprints. The goal is to ensure the system is functional, testable, and provides immediate value to students and staff with minimal complexity.

3. Prioritization Technique

We applied the MoSCoW prioritization method to rank all Product Backlog Items. This technique provides a clear and structured way to communicate priorities with stakeholders while ensuring the team focuses on delivering maximum value within the sprint constraints.

3.1. MoSCoW Categories

Category	Meaning	Applied To
Must Have	Core functionality required for a usable MVP. System cannot launch without these.	PBI-1, PBI-2, PBI-3, PBI-4
Should Have	Important but not critical for the first increment. High business value.	PBI-5, PBI-6
Could Have	Nice-to-have enhancements. Included only if time permits.	PBI-7
Won't Have	Out of scope for Sprint 1. Deferred to future sprints.	PBI-8

3.2. Prioritization Criteria

Each PBI was evaluated against four criteria before assigning its MoSCoW category:

- Business Value: Impact on university operations and user experience
- Dependencies: Some features require others to function (e.g., login must exist before profile)
- Technical Feasibility: What can realistically be delivered within one sprint
- MVP Usability: The system must be minimally functional and usable after Sprint 1

4. Product Backlog Items (Sprint 1 MVP)

The following table summarizes all 8 PBIs, their MoSCoW priority, story points, assigned team member, and Sprint 1 scope status. Full acceptance criteria and Jira links are provided in the detailed descriptions below.

Key	PBI Name	MoSCoW	SP	Assignee	Sprint 1 Scope
SCRUM-4	User Authentication & Role Mgmt	Must Have	8	Mohamed Abosamra	Included
SCRUM-5	Basic User Profile View	Must Have	5	Basmala Hany	Included
SCRUM-6	University Announcements Dashboard	Must Have	5	Mai Hamed	Included
SCRUM-7	Course Catalog View	Must Have	5	Maryam Hamdy	Included
SCRUM-8	Basic Course Enrollment Request	Should Have	6	Mariam Shaker	Included
SCRUM-9	Staff Directory	Should Have	5	Mariam Riyad	Included
SCRUM-10	Basic Messaging Placeholder	Could Have	3	Mariam Riyad	Included
SCRUM-11	Facilities Overview Page	Won't Have	0	Unassigned	Backlog Only

4.1. PBI-1: User Authentication & Role Management (SCRUM-4)

User Story: As a user, I want to log in to the system so that I can access my account securely.

Justification (Must Have):

Essential for system access and security; all other features depend on authenticated user roles.

Acceptance Criteria:

- User enters username and password
- If credentials are correct, login is successful and session starts
- If credentials are incorrect, an error message is displayed
- User is redirected based on role: Student, Staff, or Admin
- Access is restricted based on the assigned role

Out of Scope: Password recovery, multi-factor authentication.

4.2. PBI-2: Basic User Profile View (SCRUM-5)

User Story: As a user, I want to view my profile so that I can check my personal information.

Justification (Must Have):

Core to user identity visibility and required for a minimally usable personalized system experience.

Acceptance Criteria:

- Profile page displays username and role
- Information is read-only
- Page loads correctly without errors

Out of Scope: Profile editing, profile picture upload, advanced settings.

4.3. PBI-3: University Announcements Dashboard (SCRUM-6)

User Story: As a user, I want to view university announcements so that I stay informed about important news.

Justification (Must Have):

Provides immediate value by enabling core communication across all users, a key system objective.

Acceptance Criteria:

- List of announcements is displayed
- Each announcement includes title, content, and date
- Accessible to all users regardless of role
- Page updates when new announcements are added (future improvement)

Out of Scope: Creating announcements (Admin feature, future sprint), commenting or reacting.

4.4. PBI-4: Course Catalog View (SCRUM-7)

User Story: As a student, I want to browse available courses so that I can explore my options.

Justification (Must Have):

Fundamental academic feature allowing students to explore available courses, enabling system usefulness.

Acceptance Criteria:

- Course list is displayed

- Each course includes name, code, and description
- User can scroll through courses
- System handles a large number of courses without crashing

Out of Scope: Enrollment actions, course grading or assessments.

4.5. PBI-5: Basic Course Enrollment Request (SCRUM-8)

User Story: As a student, I want to request enrollment in a course so that I can join it.

Justification (Should Have):

Important for extending functionality beyond browsing, but the system remains usable without full enrollment capability.

Acceptance Criteria:

- User can select a course from the catalog
- Enrollment request is submitted successfully
- Request status is stored as Pending
- Duplicate requests are prevented
- Confirmation message is displayed after submission

Out of Scope: Automatic approval, admin approval workflow, payment or registration fees.

4.6. PBI-6: Staff Directory (SCRUM-9)

User Story: As a user, I want to view the staff directory so that I can contact professors and TAs.

Justification (Should Have):

Enhances communication and accessibility of staff, but not critical for initial MVP functionality.

Acceptance Criteria:

- Staff list is displayed
- Each entry includes name, department, and contact info
- User can browse the directory

Out of Scope: Staff performance tracking, HR or payroll features.

4.7. PBI-7: Basic Messaging Placeholder (SCRUM-10)

User Story: As a user, I want to see a messaging section so that I am aware of future communication features.

Justification (Could Have):

Low implementation effort and helps communicate future system vision, but delivers minimal immediate value.

Acceptance Criteria:

- Basic UI placeholder for messaging section is visible
- No real message transmission required in Sprint 1

Out of Scope: Real-time chat, notifications, message storage system.

4.8. PBI-8: Facilities Overview Page (SCRUM-11) — Won't Have

Justification (Won't Have):

Deferred due to lower priority and higher complexity, not required for delivering a functional MVP in Sprint 1.

Deferred features include: room booking system, lab scheduling, and resource allocation.

5. Story Splitting

Large user stories were broken down into smaller, independently deliverable stories using a combination of splitting techniques. This ensures each story fits within a sprint, has clear acceptance criteria, and can be tested independently.

5.1. Splitting Techniques Used

Technique 1: Functional Decomposition

Breaking a large story into distinct features or capabilities within the same domain.

Example — Original: "As a student, I want to manage my courses so that I can enroll and track academic progress."

- As a student, I want to browse available courses so that I can explore my options. (PBI-4)
- As a student, I want to request enrollment in a course so that I can join it. (PBI-5)

Technique 2: Workflow Splitting

Splitting a story by steps in a user workflow, each step becoming an independent story.

Example — Original: "As a user, I want to manage my account so that I can use the system effectively."

- As a user, I want to log in to the system securely so that I can access my account. (PBI-1)
- As a user, I want to view my profile information so that I can check my personal details. (PBI-2)

Technique 3: Simple to Complex

Delivering a basic version first, then layering in complexity in future sprints.

- Show list of courses (Sprint 1)
- Add filtering (Future Sprint)
- Add search (Future Sprint)

Technique 4: CRUD Operations

Splitting by data operation type to deliver the most critical operations first.

- Create enrollment request (Sprint 1 - PBI-5)
- View request status (Sprint 1 - PBI-5)
- Update or delete request (Future Sprint)

Technique 5: UI vs Backend

Delivering the UI layer first, adding backend logic in a later sprint.

- Basic Messaging UI placeholder (Sprint 1 - PBI-7)
- Backend messaging logic and storage (Future Sprint)

Technique 6: Business Rules

Splitting by different rules that apply to different user roles or conditions.

- Student role access (Sprint 1)
- Staff role access (Sprint 1)
- Admin role access (Sprint 1)
- Prevent duplicate enrollment requests (Sprint 1 - PBI-5)

5.2. Summary Table

Original Large Story	Split Story 1	Split Story 2	Technique Used
Manage my courses	Browse available courses (PBI-4)	Request enrollment (PBI-5)	Functional Decomposition
Manage my account	Log in securely (PBI-1)	View profile (PBI-2)	Workflow Splitting
Full course interaction	Browse courses (PBI-4)	Submit enrollment (PBI-5)	Simple to Complex
Communication features	Messaging UI placeholder (PBI-7)	Full messaging (Future)	UI vs Backend

6. Risk Analysis

During the planning of Sprint 1, the team identified several potential risks that could impact the project timeline, system quality, and overall delivery. The following table outlines the key risks along with their potential impact and mitigation strategies:

Risk	Description	Impact	Mitigation Strategy
Task Delays	Some backlog items may require more time than estimated due to complexity or inaccurate estimation	Delays in achieving the sprint goal	Break down tasks into smaller stories and track progress through daily scrums
Integration Challenges	Difficulty in integrating different modules such as authentication with other system features	System instability or incomplete functionality	Apply modular design and perform early integration testing
Time Constraints	Limited time available to complete all planned MVP features within a single sprint	Partial delivery of MVP features	Prioritize "Must Have" items using MoSCoW and defer lower priority items
Team Coordination Issues	Miscommunication or lack of alignment between team members	Reduced productivity and increased errors	Maintain regular communication through daily meetings and Jira updates
Technical Learning Curve	Some team members may need time to adapt to new tools or technologies	Slower progress during the sprint	Encourage knowledge sharing and allocate time for quick learning

7. Definition of Done (DoD)

The team has agreed on the following Definition of Done. A Product Backlog Item is considered Done only when ALL of the following criteria are met:

#	Criterion	Details
1	User Story is Complete	Written in format: As a [role], I want [action], so that [benefit]. Acceptance criteria are fully defined.
2	Backlog Item is Documented	PBI includes: title, MoSCoW priority, story points, scope, out-of-scope, and splitting technique where applicable.
3	Jira is Updated	PBI is created in Jira with all fields filled. Issue status reflects current state. Epic link is set.
4	Prioritization is Justified	MoSCoW priority is justified based on business value, dependencies, technical feasibility, and MVP usability.
5	Team Review & Agreement	PBI has been reviewed and agreed upon by all team members. No unresolved conflicts or open questions remain.

This Definition of Done was agreed upon by all team members and applies to every PBI in Sprint 1. It ensures consistent quality and completeness across all backlog items before they are considered finished.

8. Jira Project Link

All Product Backlog Items described in this document have been created in Jira with full descriptions, acceptance criteria, MoSCoW labels, priority levels, epic links, and team member assignments.

Project URL:

<https://teamfifteenagile.atlassian.net>

Jira Structure:

- 3 Epics: User Access & Authentication (SCRUM-1), University Information & Browsing (SCRUM-2), Course Enrollment & Management (SCRUM-3)
- 8 Stories: SCRUM-4 through SCRUM-11, each with user story, acceptance criteria, MoSCoW label, and assignee
- Sprint 1 contains SCRUM-4 through SCRUM-10 (SCRUM-11 remains in backlog as Won't Have)

8.1. Team Assignments

Jira Key	Story	MoSCoW	Assignee
SCRUM-4	User Authentication & Role Management	Must Have	Mohamed Abosamra
SCRUM-5	Basic User Profile View	Must Have	Basmala Hany
SCRUM-6	University Announcements Dashboard	Must Have	Mai Hamed
SCRUM-7	Course Catalog View	Must Have	Maryam Hamdy
SCRUM-8	Basic Course Enrollment Request	Should Have	Mariam Shaker
SCRUM-9	Staff Directory	Should Have	Mariam Riyad
SCRUM-10	Basic Messaging Placeholder	Could Have	Mariam Riyad
SCRUM-11	Facilities Overview Page	Won't Have	Backlog Only